

SIMATS ENGINEERING
ASSESSMENT TOOL 3 : Scenario-Based

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA15
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COURSE OUTCOME COVERED

CO2: Analyze virtualization architectures to identify performance and security issues in

cloud computing environments. (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)

Question Count: 5

Total Marks: 25

Code of Conduct

I ____R. AKSHAYA(192524118)____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _R. Akshaya_____

Assessment Tasks

Analytical Problem Solving

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how

cloud auto-scaling and load balancing can handle this demand efficiently. How does

this approach reduce downtime and improve user experience? What role do monitoring

tools play in this setup? Relate your answer to a real-world e-commerce scenario.

2. An organization wants to store large volumes of backup data securely in the cloud.

Discuss how cloud storage solutions ensure durability and data redundancy.

How can encryption and access control protect sensitive backup data?

Explain the cost advantages compared to traditional storage systems.

Provide an example of a disaster recovery use case.

3. A mobile app company plans to deploy its application globally. Explain how content

delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support

global availability. Relate this to a real-time streaming or gaming application.

4. A financial institution requires strict compliance and data security in cloud usage.

Explain how cloud providers support regulatory compliance and auditing. What role do

logging, monitoring, and encryption play in security? How can the company ensure

data sovereignty requirements are met? Give an example involving sensitive financial

transactions.

5. A development team wants to adopt containerization for application deployment.

Explain how containers differ from virtual machines in terms of performance and

resource usage. How does orchestration (like Kubernetes) simplify deployment and

scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.
Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and	Generally clear	Somewhat unclear	Difficult to follow

		technically precise			
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Total Marks Awarded:

Assessment Tasks

Analytical Problem Solving

1. A company experiences unpredictable traffic spikes during seasonal sales. Explain how

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this approach reduce downtime and improve user experience? What role do monitoring

tools play in this setup? Relate your answer to a real-world e-commerce scenario.

Cloud Auto-Scaling

- Auto-scaling automatically increases or decreases the number of servers based on traffic.
- During seasonal sales, more servers are added to handle increased customer requests.
- When traffic reduces, extra servers are removed to save costs.

Load Balancing

- Load balancing distributes incoming user requests evenly across multiple servers.
- It prevents any single server from becoming overloaded.
- This ensures all servers are utilized efficiently.

Reducing Downtime and Improving User Experience

- Prevents website crashes during heavy traffic.
- Maintains fast response times for users.
- Ensures continuous service availability even during peak demand.
- Improves customer satisfaction and reduces cart abandonment.

Role of Monitoring Tools

- Monitor CPU, memory, network usage, and application performance.

- Detect sudden traffic spikes.
- Trigger auto-scaling policies automatically.
- Send alerts if any server or application fails.

Real-World Example

- During **Amazon Prime Day** or **Flipkart Big Billion Days**, millions of users visit simultaneously.
- Auto-scaling launches additional cloud servers.
- Load balancers distribute customer requests across these servers.
- Monitoring tools continuously track performance, ensuring smooth shopping without downtime.

2. An organization wants to store large volumes of backup data securely in the cloud.

Discuss how cloud storage solutions ensure durability and data redundancy.

Durability and Data Redundancy

- Cloud providers store multiple copies of backup data across different locations.
- If one storage device or data center fails, another copy remains available.
- This ensures high durability and data availability.

Encryption and Access Control

- Encryption protects data while it is stored and during transmission.
- Access control allows only authorized users to access backup files.
- Multi-factor authentication (MFA) provides additional security.
- Role-based permissions prevent unauthorized access.

Cost Advantages

- No need to purchase expensive storage hardware.
- Pay only for the storage actually used.
- Reduces maintenance and electricity costs.
- Automatic backups eliminate manual management efforts.

Disaster Recovery Example

- A company's primary data center is damaged due to flooding.

- Backup data stored in the cloud is quickly restored.
- Business operations continue with minimal downtime.
- Important customer and business records remain safe.

3. A mobile app company plans to deploy its application globally. Explain how content

delivery networks (CDNs) improve performance and reduce latency. How do edge

servers enhance user experience across regions? Discuss how cloud providers support

global availability. Relate this to a real-time streaming or gaming application.

How CDNs Improve Performance

- CDNs store copies of website content in multiple global locations.
- Users receive content from the nearest server instead of the main server.
- This reduces loading time and network delay.

Role of Edge Servers

- Edge servers are located close to users.
- They deliver cached content quickly.
- Reduce the distance data must travel.
- Improve application responsiveness.

Cloud Provider Support

- Cloud providers maintain data centers worldwide.
- Automatically route users to the nearest available server.
- Provide high availability even if one region experiences failures.
- Enable global application deployment.

Real-Time Streaming/Gaming Example

- **Netflix** stores video content on CDN edge servers worldwide.
- Users stream movies from nearby servers, reducing buffering.

- Online multiplayer games use CDNs to reduce latency, providing smoother gameplay for players in different countries.

4. A financial institution requires strict compliance and data security in cloud usage.

Explain how cloud providers support regulatory compliance and auditing. What role do

logging, monitoring, and encryption play in security? How can the company ensure

data sovereignty requirements are met? Give an example involving sensitive financial

transactions.

Regulatory Compliance

- Cloud providers support standards such as GDPR, ISO 27001, PCI DSS, and HIPAA.
- They provide compliance reports and certifications.
- Help organizations meet legal and industry security requirements.

Role of Logging, Monitoring, and Encryption

- Logging records all user activities and system events.
- Monitoring detects suspicious behavior and security threats.
- Encryption protects sensitive financial information both during storage and transmission.
- Audit logs support investigations and compliance verification.

Ensuring Data Sovereignty

- Organizations choose cloud regions where data is stored.
- Sensitive customer data remains within the required country.
- Providers offer region-specific storage options to meet government regulations.

Financial Transaction Example

- A bank processes online payments through cloud services.
- All transactions are encrypted.
- Every access and transaction is logged for auditing.
- Customer financial records remain stored within the country's approved data center to meet regulatory requirements.

5. A development team wants to adopt containerization for application deployment.

Explain how containers differ from virtual machines in terms of performance and

resource usage. How does orchestration (like Kubernetes) simplify deployment and

scaling? Discuss portability benefits across different cloud environments.

Provide an example of a microservices-based application.
Analytical Problem Solving Rubric

Containers	Virtual Machines
Share the host operating system	Each VM has its own operating system
Lightweight	Heavier due to full OS
Start within seconds	Take longer to start
Use fewer system resources	Consume more CPU, RAM, and storage
Higher performance	Slightly lower performance

Role of Kubernetes

- Automatically deploys containers.
- Scales applications based on user demand.
- Restarts failed containers automatically.
- Balances traffic among running containers.
- Simplifies application management.

Portability Across Clouds

- Containers package the application along with all dependencies.
- The same container runs consistently on different cloud providers.
- Reduces compatibility issues during migration.
- Supports hybrid and multi-cloud deployments.

Microservices Example

An **online shopping application** can be divided into separate microservices:

- User Authentication Service
- Product Catalog Service
- Shopping Cart Service
- Payment Service
- Order Management Service

Each service runs inside its own container and is managed by Kubernetes. If the Payment Service experiences heavy traffic, Kubernetes scales only that service without affecting the others, improving performance and resource efficiency.